Does Fusing Deformation and Shear Improve Tactile Material Classification? An Ablation Study under Contact-Condition Variability

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***Abstract*—Tactile sensing lets a robot identify what it touches from the mechanical response of the contact, independent of vision. We study whether *fusing* the two channels of a vision-based tactile sensor—surface deformation and shear—improves material classification over either channel alone, and how robustly that advantage holds as contact conditions vary. Using a Daimon WS sensor we collected 10 materials under two contact modes (a controlled *press* and a more variable *airhold* squeeze), and compared classical classifiers and deep temporal models under a leakage-free, validation-based evaluation. Three findings emerge. First, the *spatial* structure of the contact footprint dominates raw force magnitude (65.5% vs. 42.8% accuracy), confirming that material identity is encoded geometrically. Second, fusing deformation and shear beats the best single modality by 4–9 percentage points in every condition, because compliant and textured materials activate the shear channel. Third, a temporal self-attention model raises honest held-out accuracy from a ∼71% classical ceiling to ∼80%, while a controlled nine-configuration regularization sweep shows the residual train–test gap is driven by data quantity rather than model capacity. Visually identical PLA cubes of differing stiffness are separated by touch with near-zero cross-confusion. We report a reproducibility-bounded result of ∼80% ± 5–6% on 10 materials and discuss the hand-collection force confound as the principal limitation.**

***Index Terms*—tactile sensing, material classification, deformation, shear, sensor fusion, temporal attention, robotic manipulation**

I. Introduction and Research Question

Touch gives a robot information vision cannot: the mechanical response of a contact reveals stiffness, texture, and compliance even when two objects look identical. Vision-based tactile sensors expose this response as two physically distinct channels—*deformation* (the normal/vertical displacement of a soft sensing surface, encoding contact geometry) and *shear* (the lateral/tangential displacement, proportional to tangential force) [1]. A natural question for any fusion-based perception system is whether using both channels is worth the cost over using one.

Following the PICO formulation for robotics research questions, we phrase our study as: *Does fusing deformation and shear (intervention) improve material-classification accuracy (outcome) on a set of everyday rigid, compliant, and textured objects (population/context) compared to using either modality alone (baseline), and how does that advantage change as contact-condition variability increases?* This is testable, measurable (classification accuracy under cross-validation), controlled (same sensor, materials, and pipeline across conditions), and non-trivial (prior work has not run the controlled single-vs-fused comparison on this modality).

We formalize three hypotheses:

- **H1**: fusion outperforms the best single modality at low contact variability;
- **H2**: the fused advantage persists, and is not eliminated, as contact variability grows;
- **H3**: each single modality fails on a physically predictable subset of materials, which fusion covers.

Variables. The *independent* variables are the sensing modality (deformation-only / shear-only / fused), the feature representation (magnitude / shape / spatial), the model class (classical / temporal deep), and the contact condition (press / airhold). The *dependent* variable is 5-fold cross-validated classification accuracy (with per-class confusion and the train–test gap as secondary metrics). *Control* variables are the sensor, the material set, the trial count, and the evaluation protocol, held fixed across comparisons.

Contributions. (i) A controlled feature-group and modality ablation that isolates the contribution of spatial structure and of shear; (ii) a demonstration that temporal self-attention—not added model capacity—breaks the classical accuracy ceiling; (iii) an honest, reproducibility-bounded evaluation with a regularization sweep that diagnoses the residual generalization gap as data-limited; and (iv) a clear "touch-beats-vision" result on visually identical objects.

II. Literature Analysis

Our study sits on a ladder of four works spanning sensing principles to the current state of the art.

Foundation. GelSight [1] established the vision-based tactile paradigm: a camera images a soft elastomer whose vertical and lateral surface deformation encodes contact geometry, with marker-motion magnitude "roughly proportional to" contact force and shear. It grounds our treatment of deformation and shear as the two physically meaningful channels and motivates modeling the temporal loading dynamics of a soft, viscoelastic contact.

Direct lab/pipeline precedent. Amin, Gianoglio and Valle [2], from the same institution on the same Baxter platform, perform embedded real-time object *hardness* classification, comparing an SVM, single-layer feed-forward networks, and a CNN (CNN best, >98%). Crucially their sensor is a *PVDF piezoelectric pressure array*—not a deformation+shear field—and their features are the mean and standard deviation of each tactile frame. Our work mirrors their Baxter-mounted classical-plus-deep pipeline but differs in modality and, importantly, shows that the per-frame mean/std pooling they adopt discards exactly the spatial structure that we find carries material identity. (We note for accuracy that the "170 trials" in [2] refer to materials-machine compression ground-truthing, not tactile-dataset size.)

Sensor-specific reference. Liao et al. [3] use the Daimon DM-Tac sensor—the same vendor family as ours—and decompose contact into normal and shear components, estimating fruit hardness from the *dynamics of the normal force* ($\mathrm{d}F_z/\mathrm{d}t$, $\mathrm{d}F_z/\mathrm{d}\delta$). This validates our sensor lineage and supports modeling temporal loading dynamics rather than a single peak frame. Notably, [3] *decomposes shear but uses only the normal force*, motivating precisely the fusion question we test.

State-of-the-art ceiling. FG-CLTP [4] is a fine-grained contrastive language–tactile pretraining framework that aligns 3D tactile point clouds with quantitative contact-state descriptions, reporting 95.9% accuracy on a 100k-pair dataset. Its central argument—that *quantitative* contact states (force magnitude, contact geometry) carry more than qualitative texture descriptors, and that spatial 3D representations outperform image-only ones—directly supports our premise. It also delimits the resource frontier (data scale, 3D representation, pretraining) that our small single-sensor study does not reach, framing our future work.

Positioning. Among these, ours is the only study that (a) uses deformation+shear as its explicit fused modality, (b) runs the controlled deformation/shear/fused ablation that none of the four perform, and (c) reports under an honest, reproducibility-bounded protocol. Where [2] averages the spatial field away and [3] discards shear, our contribution is to quantify what is recovered by keeping both.

III. Methodological Proposal

A. Sensor and signals

We use a Daimon WS vision-based tactile sensor providing, per frame, a deformation field and a shear field, each a two-component vector map downsampled to 30×40. A trial is a temporal stack of such frames spanning approach, loading, hold, and release. Concatenating the two modalities yields a $(T, 30, 40, 4)$ tensor per trial.

B. Feature representations (classical)

We extract a 30-dimensional feature set in four groups: *magnitude* (peak, mean, std of the field), *shape* (loading-curve descriptors), *spatial-deformation* (centroid, spread, area, concentration of the deformation footprint), and *spatial-shear* (the same for shear). The feature-group split lets us ablate which physical aspect of the contact carries identity.

C. Models

Classical: SVM (RBF), Random Forest, and XGBoost on the 30 features. *Deep, temporal*: a CNN encodes each frame; a bidirectional LSTM models the loading sequence; a temporal self-attention layer re-weights informative frames before a linear classifier. We additionally test a dual-branch variant that processes deformation and shear in separate streams, to isolate whether explicit modality separation helps.

D. Honest evaluation protocol

All accuracies are 5-fold cross-validated. For the deep models, early stopping uses an *internal validation split carved from the training fold*; the test fold is never used for model selection. This removes the optimistic bias of selecting the epoch by test-fold accuracy and yields reviewer-proof numbers. We report mean accuracy, standard deviation across folds, the train–test gap, and the aggregated confusion matrix.

IV. Experiment Design

A. Materials and contact conditions

We collected 10 materials spanning rigid, compliant, and textured classes: carton box, ceramic cup, hardwood, two visually identical PLA cubes of differing stiffness (`pla_hard`, `pla_soft`), two textured plastics, rubber, steel cup, and a wallet. Each material was sampled under two contact conditions, which constitute our variability axis: *press* (a controlled, consistent contact) and *airhold* (the object squeezed against the sensor in the air, producing more variable contact and stronger shear). Twenty trials per material per condition were collected.

B. Collection constraint and its consequence

Robot access was limited, so data were hand-collected with a release-gated protocol guaranteeing one clean press per trial. This makes the applied force *uncontrolled*, so force is partially confounded with material identity. This is the principal limitation of the study and is treated explicitly: results are validation-grade rather than force-controlled-deployment-grade. The two contact conditions are used precisely to probe robustness to this variability (H2).

C. Baseline

Per the RT2 methodology, the baseline is the "most common solution": a classical classifier on magnitude features (the mean/std-style representation of [2]) and a standard CNN-LSTM without temporal attention. The intervention (spatial features; modality fusion; temporal attention) is compared against these.

V. Results and Visualization

A. Spatial structure dominates magnitude

A feature-group ablation (Table I) shows spatial features reach 65.5% on the pooled set versus 42.8% for magnitude—a $\sim 1.5\times$ advantage, robust across conditions. Exploratory analysis (Fig. 1) explains why: raw loading curves separate materials by force, but after per-trial normalization the curves partially collapse, leaving residual *shape* differences as the

TABLE I
FEATURE-GROUP ABLATION (5-FOLD CV ACCURACY, SVM).

Group (dim)	press	airhold	all
Magnitude (7)	50.5	44.0	42.8
Shape (5)	27.0	33.5	26.2
Spatial (18)	**68.5**	**67.0**	**65.5**
ALL (30)	74.5	72.5	71.0

TABLE II
MODALITY ABLATION (5-FOLD CV ACCURACY, SVM).

Modality	press	airhold	all
Deformation only	67.5	68.0	62.0
Shear only	65.5	62.5	55.5
Fused	**74.5**	**72.5**	**71.0**
Gain over best single	+7.0	+4.5	+9.0

discriminative signal. Per-material contact imprints (Fig. 2) are visibly distinct, confirming that identity is encoded geometrically.

B. Fusion beats single modality (answer to the research question)

The modality ablation (Table II, Fig. 3) confirms H1: fusing deformation and shear beats the best single modality by 4–9 points in every condition. H3 holds in the confusion structure: deformation struggles where stiffness is similar, shear is weak on smooth rigid items, and fusion covers both. Feature-importance analysis ranks the spatial spread of *both* fields highest, and the shear channel—negligible on rigid pilot materials— becomes informative once compliant and textured materials are present (supporting H2).

C. Temporal attention breaks the ceiling

Classical models and a baseline CNN-LSTM converge near 71%. Temporal self-attention raises *honest* held-out accuracy to $\sim$80% (press 80.5%, airhold 79.5%, pooled 79.2%). A dual-branch variant matches but does not exceed single-stream attention, isolating temporal attention—not added capacity or explicit modality separation—as the source of the gain (Fig. 4).

D. The gap is data-limited (regularization sweep)

A nine-configuration sweep (Fig. 5) finds that only gentle output regularization helps: label smoothing gives the best single result (82.8%), weight decay helps marginally, while capacity reduction, temporal cropping, and augmentation are neutral or harmful. The train–test gap stays at +12–17% under every intervention, and two runs of the identical architecture scored 82.8% and 77.0%—a $\sim$6-point swing from random seed alone. We therefore report a reproducibility-bounded **$\sim$80% $\pm$ 5–6%** on 10 materials, with 82.8% as the best single configuration. The immovable gap is direct evidence that data quantity ($\sim$20 trials/class), not model capacity, is the binding constraint.

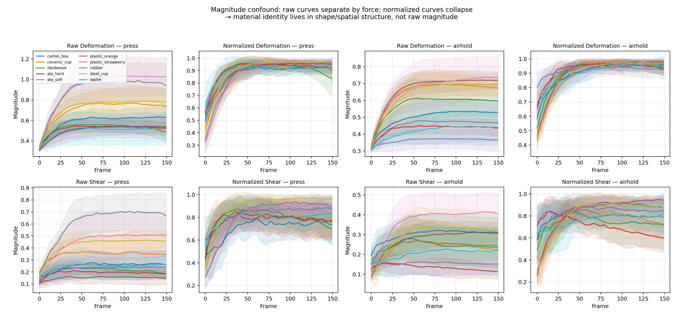

Fig. 1. Raw loading curves separate materials by force; normalized curves partially collapse, leaving shape as the discriminative signal.

E. Touch beats vision

The two PLA cubes are visually indistinguishable yet differ in stiffness. In the honest confusion matrix (Fig. 6) they show near-zero cross-confusion—direct evidence that the mechanical-response modality discriminates what a camera cannot. Residual errors concentrate in physically similar pairs (the rigid-flat cluster; the two cups), confirming the classifier's mistakes track genuine material similarity.

VI. DISCUSSION AND LIMITATIONS

Answer to the research question. Fusing deformation and shear improves material classification over either modality alone by 4–9 points (H1), the advantage persists across both contact conditions and is driven by compliant and textured materials activating shear (H2, H3). The underlying signal is *spatial*: contact-footprint geometry, not raw force magnitude.

What attention adds, and what it does not. Temporal self-attention recovers $\sim$9 points over the classical/baseline ceiling, showing the limitation there was partly temporal-modeling, not purely data. But the nine-configuration sweep shows the residual generalization gap is *data-quantity-driven*: it is immovable by capacity reduction or augmentation, and only gentle output regularization helps marginally.

Limitations. (1) *Hand collection* leaves applied force confounded with material identity; this caps accuracy and is the dominant limitation. (2) *Small data* ($\sim$20 trials/class) yields wide fold variance ($\pm$5–6%) and a $\sim$14% train–test gap, motivating band-based reporting. (3) *Tactile-specific augmentation constraint*: because the label is encoded in spatial geometry and magnitude, standard image augmentations (rotation, magnitude scaling, even $\pm$1 px translation) corrupt the label and degrade accuracy—only temporal jitter and small sensor-scale noise are label-preserving. (4) *Offline, single-sensor, single-operator*: no inter-session or on-robot deployment results yet.

Future work. Constant-force robotic collection (removing the confound), more trials per class (closing the gap), an on-robot deployment/transfer test, and richer 3D representations or contrastive pretraining in the direction of [4].

Conclusion. On 10 materials with a deformation+shear tactile sensor, fusing the two modalities and modeling the temporal loading dynamics with attention yields $\sim$80% $\pm$ 5–6% honest accuracy, separating even visually identical objects

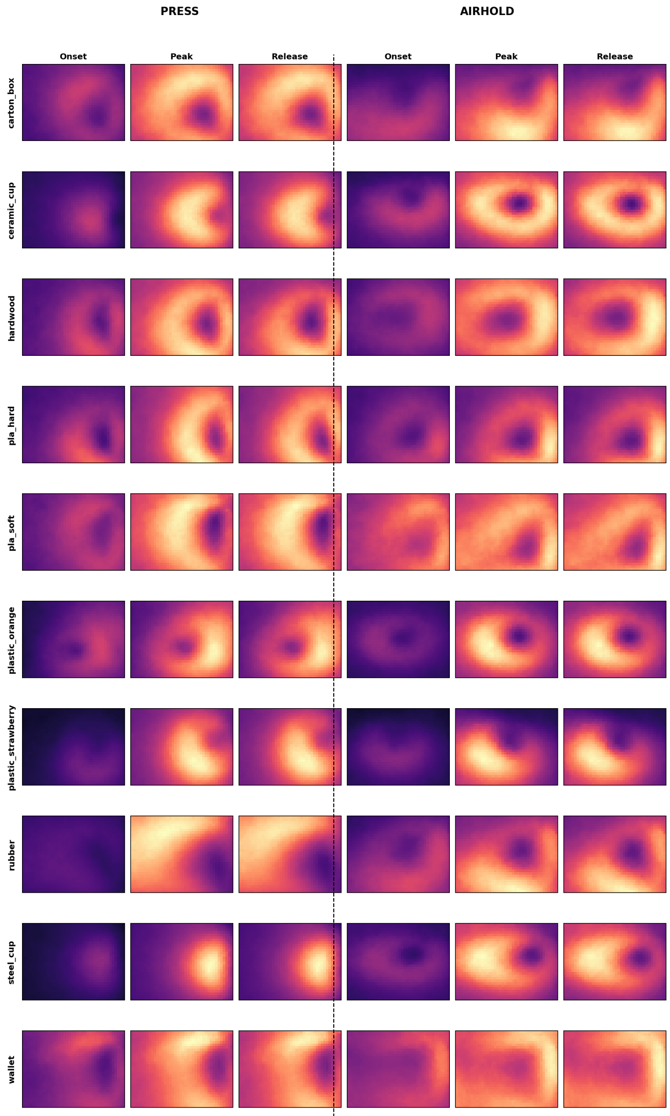

Fig. 2. Per-material contact imprints (onset/peak/release; press and airhold). Each material leaves a distinct spatial footprint.

by touch. The result is a complete, self-aware answer to the research question, bounded by data scale rather than by method.

Reproducibility

Code, per-fold logs, and figure-generation scripts are in the project repository. Figures are regenerated from the dataset and result files; all reported numbers use the honest validation-based protocol.

References

[1] W. Yuan, S. Dong, and E. H. Adelson, “GelSight: High-Resolution Robot Tactile Sensors for Estimating Geometry and Force,” *Sensors*, vol. 17, no. 12, 2762, 2017.

[2] Y. Amin, C. Gianoglio, and M. Valle, “Embedded real-time objects’ hardness classification for robotic grippers,” *Future Generation Computer Systems*, vol. 148, pp. 211–224, 2023.

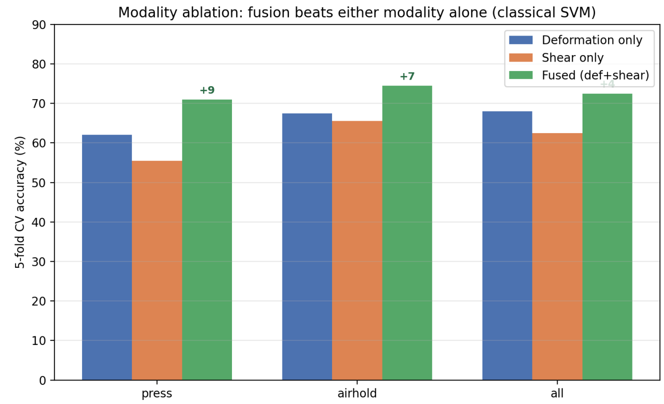

Fig. 3. Modality ablation: fused deformation+shear beats either alone in every condition (annotated gain over best single modality).

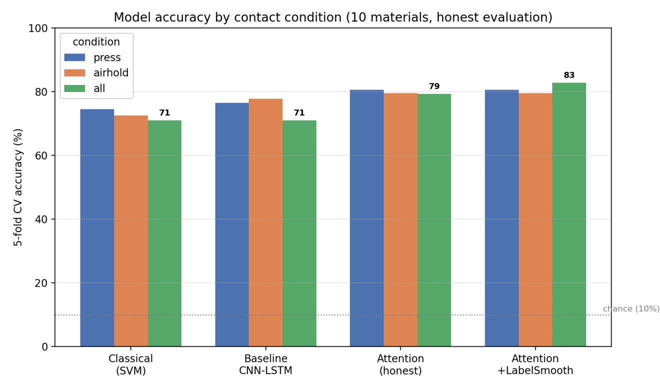

Fig. 4. Accuracy by model and condition under honest evaluation. Attention breaks the ∼71% classical ceiling; label smoothing is best.

[3] Z. Liao, Y. Du, J. Duan, H. Liang, and M. Y. Wang, “Quantitative Hardness Assessment with Vision-based Tactile Sensing for Fruit Classification and Grasping,” 2025.

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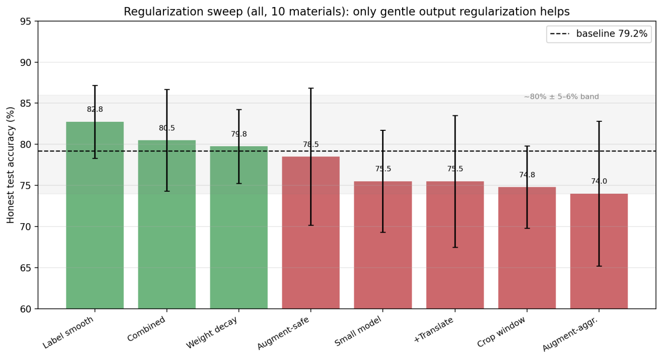

Fig. 5. Regularization sweep (10 materials, honest eval). Only gentle output regularization helps; the ∼80% ± 5–6% band is shaded.

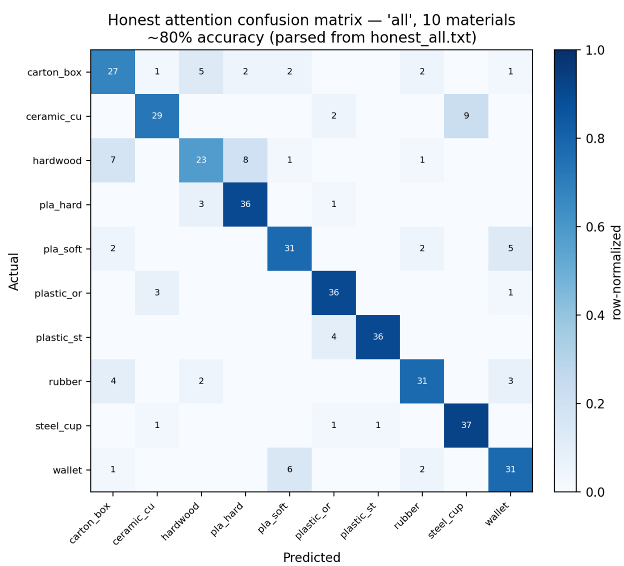

Fig. 6. Honest attention confusion matrix (pooled, 10 materials). The PLA pair shows near-zero cross-confusion—"touch beats vision".